Mastery in Data Analytics and Visualizations and Career Advancement

Lab sheet 07

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import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

df = pd.read_csv("Mall_Customers.csv")
X = df[["Annual Income (k$)", "Spending Score (1-100)"]]

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

inertias = []
sil_scores = []
ks = range(2, 11)

for k in ks:
    km = KMeans(n_clusters=k, random_state=42, n_init=10)
    labels = km.fit_predict(X_scaled)
    inertias.append(km.inertia_)
    sil_scores.append(silhouette_score(X_scaled, labels))
```

```
plt.plot(list(ks), inertias, marker="o")
plt.xlabel("K")
plt.ylabel("Inertia")
plt.title("Elbow Method")
plt.show()

plt.plot(list(ks), sil_scores, marker="o")
plt.xlabel("K")
plt.ylabel("Silhouette Score")
plt.title("Silhouette Score")
plt.show()

k = 5
model = KMeans(n_clusters=k, random_state=42, n_init=10)
df["Cluster"] = model.fit_predict(X_scaled)

plt.scatter(df["Annual Income (k$)"], df["Spending Score (1-100)"], c=df["Cluster"])
plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score (1-100)")
plt.title("Customer Segments using K-Means")
plt.show()

print(df.groupby("Cluster")[["Annual Income (k$)", "Spending Score (1-100)"]].mean())

# Questions and Answers
# Q1. What is clustering? How does it differ from classification?
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Clustering groups similar observations without predefined labels, while classification learns from labeled data to assign observations to known classes.

Q2. Explain the working principle of the K-Means Clustering algorithm.

K-Means chooses K centroids, assigns each observation to its nearest centroid, recalculates centroids from the assigned points, and repeats until the assignments stabilize or the stopping condition is reached.

Q3. Why is feature scaling important before applying K-Means clustering?

K-Means uses distance calculations. Features with larger numerical scales can dominate those with smaller scales, so standardization or normalization helps each feature contribute more fairly.

Q4. What is the Elbow Method, and how does it help determine the optimal number of clusters?

The Elbow Method plots within-cluster sum of squares (inertia) for different K values. The point where the improvement starts becoming much smaller is used as a practical choice for K.

Q5. What is the Silhouette Score? How is it used to evaluate clustering performance?

The Silhouette Score measures how close points are to their own cluster compared with other clusters. Higher values generally indicate better-separated and more cohesive clusters.

Q6. Why is K-Means considered an unsupervised machine learning algorithm?

K-Means works without a target label supplied in advance. It discovers groups from the structure of the input data.

Q7. Mention any four real-world applications of customer segmentation.

Targeted marketing, personalized product recommendations, customer loyalty programs, and differentiated pricing or offers.

Q8. What are the limitations of the K-Means algorithm?

The number of clusters K must be selected, results can depend on initialization, it is sensitive to scale and outliers, and it works best when clusters are reasonably compact and separated.

Q9. How can businesses use customer segmentation to improve marketing and customer retention?

Businesses can identify groups with different needs and spending patterns, then create targeted offers, personalized communication and retention strategies for each segment.

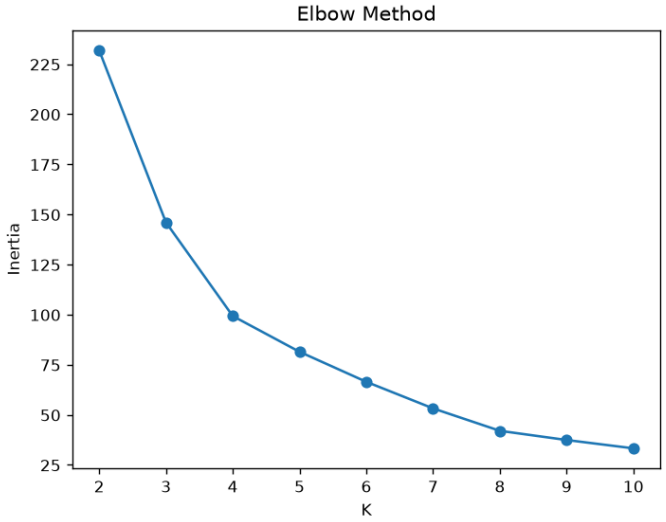
Q10. Compare K-Means Clustering with Hierarchical Clustering based on their working principles and applications.

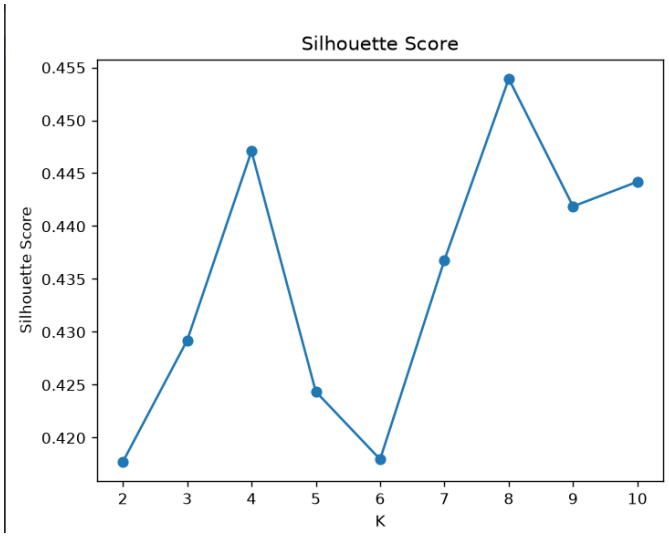
K-Means directly partitions data into a chosen number of clusters by repeatedly updating centroids. Hierarchical clustering builds a hierarchy of nested clusters that can be viewed as a dendrogram. K-Means is often efficient for larger datasets, while hierarchical clustering is useful when the nested relationships among groups are important.

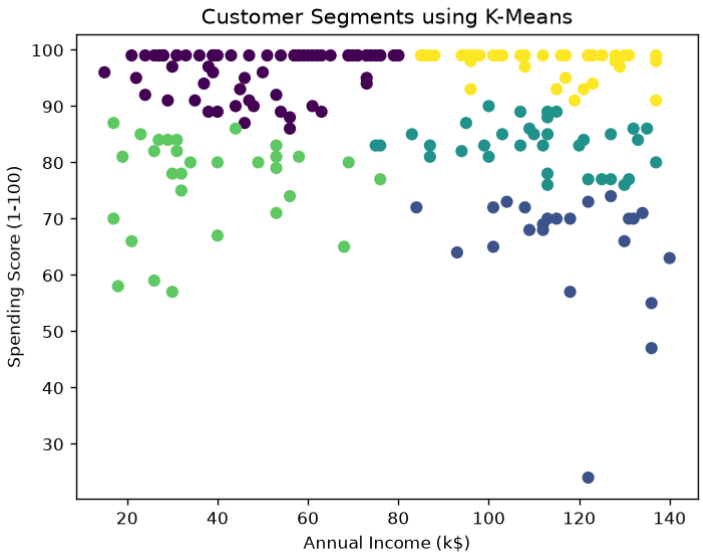
OUTPUT

PS D:\PROGRAMS\Career advancement lab\lab sheet 7> python -u "d:\PROGRAMS\Career advancement lab\lab sheet 7\Experiment_7_code.py"

	Annual Income (k$)	Spending Score (1-100)
Cluster		
0	50.929577	96.436620
1	117.304348	65.347826
2	110.705882	83.117647
3	38.500000	76.466667
4	111.642857	97.809524


Elbow Method
Inertia
K
225
200
175
150
125
100
75
50
25
2
3
4
5
6
7
8
9
10


Silhouette Score
Silhouette Score
K
0.455
0.450
0.445
0.440
0.435
0.430
0.425
0.420
2
3
4
5
6
7
8
9
10


Customer Segments using K-Means
Spending Score (1-100)
Annual Income (k$)
100
90
80
70
60
50
40
30
20
40
60
80
100
120
140